

Exact Benchmark for the Laplace AR(1) Diffusion: Structure Theorems, Scalar Observables, and the K=2 Obstruction

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Abstract

We consolidate the exact solution of the scalar Laplace AR(1) model under Ornstein–Uhlenbeck (OU) corruption into a theorem-structured benchmark. Four structure theorems (Factorisation, Score, Curvature, Fourier) express the $K=1$ joint noisy density as a Gaussian site mode multiplied by a one-dimensional non-Gaussian bond whose geometry is controlled by a single scalar function $\kappa_t(r)$. From this representation we derive closed-form scalar observables that replace the Gaussian precision-matrix diagnostics of the AR(1) benchmark: in particular, the dimensionless centre curvature $\kappa_t(0)\tilde{b}^2 = 1/(a R(a)) - 1$, expressed through Mills’ ratio $R(a) = \Phi(-a)/\phi(a)$ of the argument $a = \tau/\tilde{b}$. An exact attempt at the $K=2$ case reveals a sharp obstruction: the Gaussian sector in innovation-frequency coordinates has non-vanishing cross-covariance $-\alpha\Delta$ between the two innovation channels, so the product factorisation into two independent 1D bonds fails. The density nevertheless admits an exact representation as a sum of four terms involving bivariate normal CDFs, and a diagonal rational sector in Fourier space provides a reliable numerical route. All closed-form formulas are numerically validated against quadrature, Fourier inversion, and Monte Carlo to machine precision.

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1 Setting and notation

We study the scalar AR(1) chain with Laplace innovations, noised through an Ornstein–Uhlenbeck (OU) channel. For $k \in \{0, 1, 2, \dots\}$,

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad A_{k+1} = \alpha A_k + \varepsilon_k, \quad \varepsilon_k \sim \text{Lap}(0, b) \text{ i.i.d.}, \quad (1)$$

$$X_k = \beta A_k + \sqrt{\Delta} Z_k, \quad \beta = e^{-t}, \quad \Delta = 1 - e^{-2t}, \quad Z_k \sim \mathcal{N}(0, 1) \text{ i.i.d.} \quad (2)$$

Throughout we use the structural constants of Table 1, all defined as in the derivation note [1].

Symbol	Meaning	Definition
β	OU shrinkage	e^{-t}
Δ	OU broadening variance	$1 - \beta^2$
v	variance of X_0	$\beta^2 \sigma_0^2 + \Delta$
s^2	posterior variance of $A_0 \mid X_0$	$\sigma_0^2 \Delta / v$
$\tilde{b}$	rescaled Laplace scale	βb
τ^2	residual Gaussian variance	$\Delta + \alpha^2 \beta^2 s^2$
ρ	AR-coefficient on x_0 in the residual	$\alpha \beta^2 \sigma_0^2 / v$
ν	residual offset	$\alpha \beta \Delta \mu_0 / v$
c	leakage of x_0 into y_1	$\alpha \Delta / v$
a	Mills' argument	$\tau / \tilde{b}$
τ_G^2	Gaussian-benchmark residual variance	$\tau^2 + 2\tilde{b}^2$

Table 1: Structural constants. All are functions of $(\alpha, \sigma_0^2, \mu_0, t)$ only; the innovation parameter b enters exclusively through $\tilde{b} = \beta b$.

The central objects of study are the noisy joint density $p_t(x_0, x_1)$ (for $K = 1$) or $p_t(x_0, x_1, x_2)$ (for $K = 2$), the *joint score* $S(x, t) = \nabla_x \log p_t(x)$, and the curvature field $H_t(x) = -\nabla_x^2 \log p_t(x)$.

Remark. We emphasise that the relevant object is the *joint* score on the full noisy sequence, not the product of per-frame marginal scores. The two are equal only under coordinatewise independence, which the AR(1) structure explicitly breaks.

2 Structure theorems for $K = 1$

The starting observation is that the clean chain (1) is Markov, and the OU channel (2) is coordinatewise Gaussian and conditionally independent. Both properties survive noising, and together they produce a one-step Markov factorisation of p_t :

$$p_t(x_0, x_1) = p_t(x_0) p_t(x_1 \mid x_0). \quad (3)$$

The two theorems below turn (3) into an explicit one-dimensional residual representation.

2.1 Factorisation and score

Theorem 1 (Factorisation, $K=1$). *Under (1)–(2), the joint noisy density factorises exactly as*

$$p_t(x_0, x_1) = \underbrace{\frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta\mu_0)^2}{2v}\right)}_{\text{Gaussian site } g_t(x_0)} \cdot \underbrace{h_t(r)}_{\text{1D non-Gaussian bond}}, \quad r := x_1 - \nu - \rho x_0, \quad (4)$$

where h_t is the Normal-Laplace density – the law of $R_t := \beta\varepsilon + G_t$ with $\varepsilon \sim \text{Lap}(0, b)$ and $G_t \sim \mathcal{N}(0, \tau^2)$ independent – given in closed form by

$$h_t(r) = \frac{e^{a^2/2}}{2\tilde{b}} \left[e^{-r/\tilde{b}} \Phi\left(\frac{r}{\tau} - a\right) + e^{+r/\tilde{b}} \Phi\left(-\frac{r}{\tau} - a\right) \right], \quad a = \tau/\tilde{b}. \quad (5)$$

Sketch. Since $(X_0, X_1) \mid A$ is conditionally independent Gaussian and $A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$, the marginal of X_0 is $\mathcal{N}(\beta\mu_0, v)$, giving $g_t(x_0)$. Conditioning on $X_0 = x_0$ yields $A_0 \mid X_0 \sim \mathcal{N}(m(x_0), s^2)$ with $m(x_0) = \mu_0 + \beta\sigma_0^2(x_0 - \beta\mu_0)/v$. Writing $A_0 = m(x_0) + \xi$ with $\xi \sim \mathcal{N}(0, s^2)$ independent of (X_0, ε, Z_1) gives $X_1 \mid X_0 = \nu + \rho x_0 + R_t$ with $R_t = \beta\varepsilon + G_t$, $G_t = \beta\alpha\xi + \sqrt{\Delta}Z_1 \sim \mathcal{N}(0, \tau^2)$. The residual density h_t is obtained by direct convolution of a Laplace and a Gaussian density; the integral splits into sign cases whose completion of squares yields (5). Full derivation in [1], §4–5. $\square$

Theorem 2 (Joint score on the residual ray). Define $\psi_t(r) := (\log h_t)'(r)$. Then

$$S(x, t) = \begin{pmatrix} -\frac{x_0 - \beta\mu_0}{v} - \rho\psi_t(r) \\ \psi_t(r) \end{pmatrix}, \quad \psi_t(r) = \frac{1}{\tilde{b}} \frac{L_-(r) - L_+(r)}{L_+(r) + L_-(r)}, \quad (6)$$

with $L_\pm(r) = e^{\mp r/\tilde{b}} \Phi(\pm r/\tau - a)$. In the tails, $\psi_t(r) \rightarrow \mp 1/\tilde{b}$ as $r \rightarrow \pm\infty$. Thus the residual score saturates, sharply distinguishing the Laplace model from the Gaussian benchmark.

2.2 Curvature on a one-dimensional manifold of matrices

Theorem 3 (Curvature, $K=1$). Let $\kappa_t(r) := -\psi_t'(r) = -(\log h_t)''(r)$. Then

$$H_t(x) = \underbrace{\begin{pmatrix} 1/v & 0 \\ 0 & 0 \end{pmatrix}}_{\text{Gaussian site Hessian}} + \kappa_t(r) \underbrace{\begin{pmatrix} \rho^2 & -\rho \\ -\rho & 1 \end{pmatrix}}_{=: M_\rho, \text{ rank}=1}. \quad (7)$$

The matrix M_ρ has eigenvalues 0 and $1 + \rho^2$, with rank-1 structure $M_\rho = nn^\top$ for $n = (\rho, -1)^\top$. Consequently, the entire two-dimensional curvature field lies on a one-parameter ray of matrices, indexed by the scalar $\kappa_t(r)$.

Remark (Strong locality, geometric). The field $H_t(x)$ is generally non-constant, so there is no literal “precision matrix.” But it is constrained to a one-dimensional affine ray in the space of symmetric matrices, and its variation is controlled entirely by one scalar function of one scalar coordinate. This is a much stronger statement than mere position dependence.

2.3 Fourier representation

Theorem 4 (Fourier, $K=1$). The characteristic function of the joint noisy law is

$$\hat{p}_t(k_0, k_1) = \exp \left[i\beta\mu_0(k_0 + \alpha k_1) - \frac{\beta^2\sigma_0^2}{2}(k_0 + \alpha k_1)^2 - \frac{\Delta}{2}(k_0^2 + k_1^2) \right] \cdot \frac{1}{1 + \tilde{b}^2 k_1^2}. \quad (8)$$

In the rotated innovation-frequency coordinates $(\ell_0, \ell_1) := (k_0 - \alpha k_1, k_1)$, the representation separates entirely:

$$\hat{q}_t(\ell_0, \ell_1) = \underbrace{\exp \left[i\beta\mu_0\ell_0 - \frac{\beta^2\sigma_0^2}{2}\ell_0^2 - \frac{\Delta}{2}((\ell_0 - \alpha\ell_1)^2 + \ell_1^2) \right]}_{\text{Gaussian: site plus linear coupling}} \cdot \underbrace{\frac{1}{1 + \tilde{b}^2\ell_1^2}}_{\text{non-Gaussian, innovation only}}. \quad (9)$$

The residual density satisfies the linear ODE

$$(1 - \tilde{b}^2 \partial_r^2) h_t(r) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{r^2}{2\tau^2}\right). \quad (10)$$

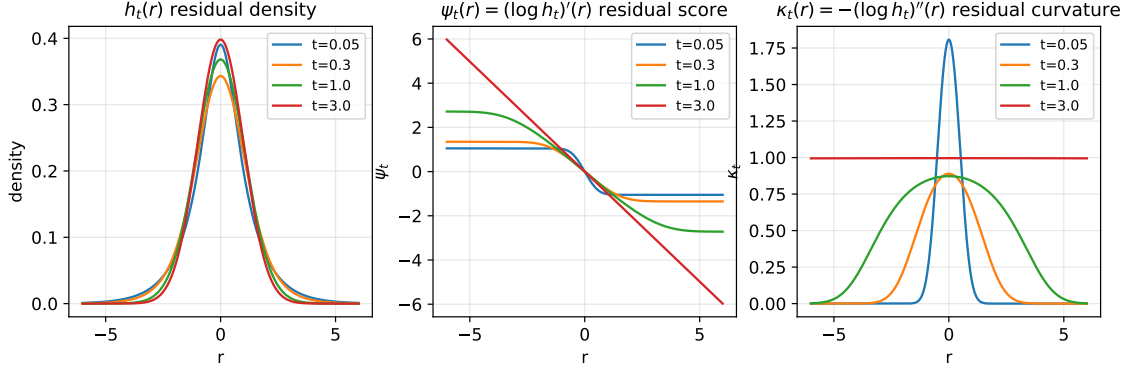


Figure 1: Residual geometry of the Laplace $K=1$ benchmark. **Left:** the Normal–Laplace density $h_t(r)$. As $t \rightarrow 0$ it concentrates near the clean Laplace cusp at $r = 0$; as $t \rightarrow \infty$ it broadens toward the Gaussian benchmark $\mathcal{N}(0, \tau_G^2)$. **Middle:** residual score $\psi_t(r)$, saturating to $\mp 1/\tilde{b}$. **Right:** residual curvature $\kappa_t(r)$, concentrated near $r = 0$ and vanishing in the tails. Parameters: $\alpha = 0.8$, $b = 1$, $\sigma_0 = 1$.

Remark (Where linearity survives). Equation (10) is a *linear* ODE in h_t ; nonlinearity enters the problem only when we pass from h_t to the score $\psi_t = h'_t/h_t$. The spectral picture is thus completely linear; geometric nonlinearity is an artefact of the logarithm.

2.4 Observed information interpretation

Proposition 5 (Meaning of H_t). *Let $B := \beta A$. Then*

$$H_t(x) = \frac{1}{\Delta} I - \frac{1}{\Delta^2} \text{Cov}(B \mid X = x), \quad \mathbb{E}_{X \sim p_t}[H_t(X)] = \mathbb{E}_{X \sim p_t}[\nabla \log p_t(X) \nabla \log p_t(X)^\top]. \quad (11)$$

Thus $H_t(x)$ is the pointwise observed information associated with the OU channel, and its p_t -expectation is the Fisher information matrix.

This identifies $H_t(x)$ as a data-space (not latent-space) object: it is controlled by *posterior covariance* about B , not by any intrinsic precision of the latent law. In the Gaussian case the two viewpoints coincide; in the Laplace case the posterior is non-Gaussian and no single precision matrix describes it.

3 Scalar observables

We now distill the residual geometry into a minimal set of scalar observables. These observables replace the Gaussian precision-matrix diagnostics (tridiagonal mass, effective bandwidth, neighbour correlation) in the non-Gaussian case; each one is a functional of the single 1D function h_t .

Definition 6 (Centre and tail curvatures). The *centre curvature* and *tail curvature* are

$$\kappa_t(0) := -(\log h_t)''(0), \quad \kappa_t(\pm\infty) := \lim_{r \rightarrow \pm\infty} \kappa_t(r). \quad (12)$$

Their difference $\Delta\kappa_t := \kappa_t(0) - \kappa_t(\infty)$ is the *centre–tail stiffness gap*.

Theorem 7 (Closed form for the centre curvature). *Let $R(a) := \Phi(-a)/\phi(a)$ denote Mills' ratio, with ϕ, Φ the standard normal pdf and cdf. Then*

$$\kappa_t(0) \tilde{b}^2 = \frac{1}{a R(a)} - 1, \quad a = \tau/\tilde{b}. \quad (13)$$

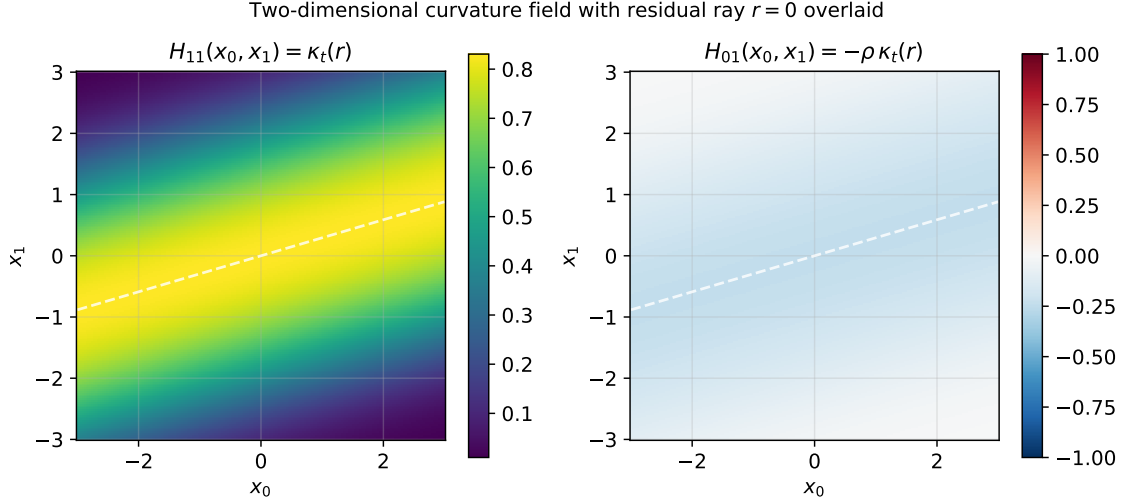


Figure 2: Two entries of the curvature field $H_t(x)$ in the (x_0, x_1) -plane. The dashed white line is the residual ray $r = 0$, i.e. $x_1 = \nu + \rho x_0$. The visible two-dimensional band structure is generated entirely by transporting the 1D profile $\kappa_t(r)$ of Figure 1 along this line (Theorem 3).

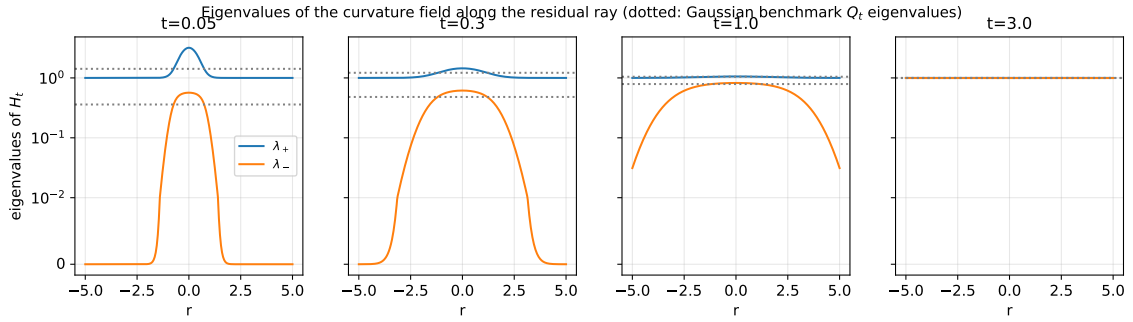


Figure 3: Eigenvalues of H_t moving along the residual ray (fixed $x_0 = 0$, varying r). The smaller eigenvalue λ_- tracks the scalar function $\kappa_t(r)$ plus a small additive shift from the Gaussian site Hessian; the larger eigenvalue λ_+ is nearly constant at $1/\nu + \kappa_t(r)(1 + \rho^2)$. Dotted lines: the constant Gaussian-benchmark eigenvalues. This figure is the geometric content of Theorem 3.

Equivalently, $\kappa_t(0) = (1/\tilde{b}^2)(1/(aR(a)) - 1)$. Moreover $\kappa_t(\pm\infty) = 0$, so $\Delta\kappa_t = \kappa_t(0)$.

Proof. By symmetry of h_t , $\psi_t(0) = 0$. From the alternative expression (note Eq. (80)) $\kappa_t(r) = \psi_t(r)^2 - 1/\tilde{b}^2 + (1/\tilde{b}^2) \cdot \phi_\tau(r)/h_t(r)$ with $\phi_\tau(r) = (2\pi\tau^2)^{-1/2}e^{-r^2/(2\tau^2)}$, evaluate at $r = 0$: $\kappa_t(0) = -1/\tilde{b}^2 + (1/\tilde{b}^2)\phi_\tau(0)/h_t(0)$. From (5), $h_t(0) = (e^{a^2/2}/\tilde{b})\Phi(-a)$. Using $\phi_\tau(0) = (\sqrt{2\pi}\tau)^{-1}$ and $\phi(a) = \phi_\tau(0) \cdot \tau \cdot e^{-a^2/2}$, $\phi_\tau(0)/h_t(0) = (\sqrt{2\pi}\tau)^{-1} \cdot \tilde{b} / (e^{a^2/2}\Phi(-a)) = 1/(aR(a))$ after substituting $a = \tau/\tilde{b}$. Then $\kappa_t(0) = (1/\tilde{b}^2)(1/(aR(a)) - 1)$.

For the tail: since $\psi_t(r) \rightarrow \mp 1/\tilde{b}$, the first two terms cancel, and $\phi_\tau(r)/h_t(r) \rightarrow 0$ because h_t has Laplace-type exponential tails dominating the faster Gaussian decay of ϕ_τ . $\square$

Remark (Limits). The identity (13) implies two sharp limits:

- $t \rightarrow 0$: $\Delta \rightarrow 0$, $\tau \rightarrow 0$, $a \rightarrow 0$, $R(0) = \sqrt{\pi/2}$, so $1/(aR(a)) \rightarrow \infty$ and $\kappa_t(0) \rightarrow \infty$ – the clean Laplace cusp.
- $t \rightarrow \infty$: $\tilde{b} \rightarrow 0$, $a \rightarrow \infty$, $R(a) \sim 1/a$, so $1/(aR(a)) \rightarrow 1$ and $\kappa_t(0)\tilde{b}^2 \rightarrow 0$: the Gaussian limit.

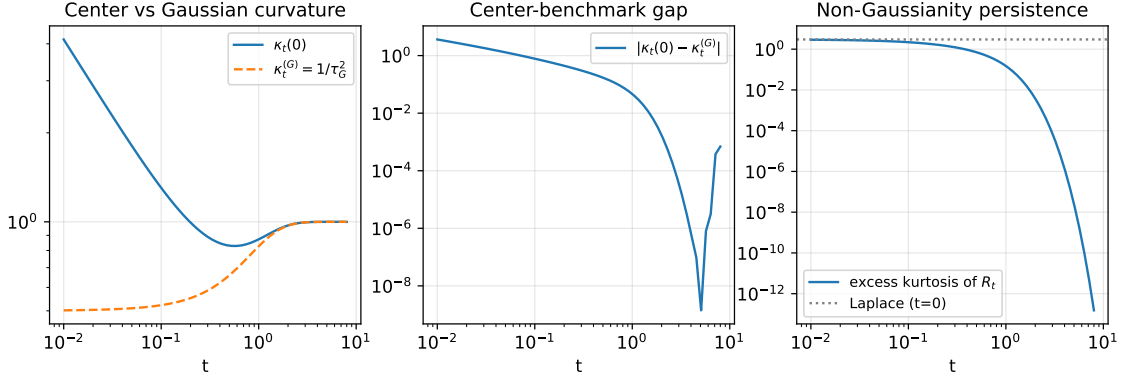


Figure 4: Scalar observables as functions of the diffusion time. **Left:** the centre curvature $\kappa_t(0)$ (solid) tracks the Gaussian-benchmark curvature $\kappa_t^{(G)}$ (dashed) in the large- t regime and diverges as $t \rightarrow 0$. **Middle:** centre-benchmark gap $|\kappa_t(0) - \kappa_t^{(G)}|$, the amount by which a Gaussian fit misrepresents the peak curvature. **Right:** excess kurtosis of R_t , a direct measure of non-Gaussianity; it interpolates monotonically from 3 at $t = 0$ (Laplace) to 0 at $t \rightarrow \infty$ (Gaussian).

Definition 8 (Gaussian benchmark). Replace the Laplace innovation by the Gaussian $\varepsilon^{(G)} \sim \mathcal{N}(0, 2b^2)$ of matching variance. The benchmark residual is Gaussian, $R_t^{(G)} \sim \mathcal{N}(0, \tau_G^2)$ with $\tau_G^2 = \tau^2 + 2b^2$, and the benchmark curvature is constant:

$$\kappa_t^{(G)} = 1/\tau_G^2 = 1/(\tilde{b}^2(a^2 + 2)). \quad (14)$$

The *centre-benchmark gap* is $\delta\kappa_t := \kappa_t(0) - \kappa_t^{(G)}$.

Observation 9 (Small- t behaviour of the gap). As $a \rightarrow 0$, $R(a) = \sqrt{\pi/2}(1 - a\sqrt{2/\pi} + O(a^2))$, so $1/(aR(a)) = \sqrt{2/\pi}/a + O(1)$, while $1/\tau_G^2 = 1/(2\tilde{b}^2)$. Therefore $\kappa_t(0) = \sqrt{2/\pi}/(a\tilde{b}^2) + O(\tilde{b}^{-2})$, while $\kappa_t^{(G)} = 1/(2\tilde{b}^2)$: the gap diverges like $1/a = \tilde{b}/\tau$ as $t \rightarrow 0$.

Definition 10 (FWHM of κ_t). The *band full-width at half-maximum* is $2r^*(t)$ where $\kappa_t(r^*) = \kappa_t(0)/2$. This is the one-dimensional analogue of the effective bandwidth $b_{0.95}(t)$ from the Gaussian benchmark.

Proposition 11 (Non-Gaussianity persistence). *The fourth cumulant of R_t is $\kappa_4(R_t) = 12\beta^4b^4$. Hence the excess kurtosis of R_t is*

$$\frac{\kappa_4(R_t)}{\text{Var}(R_t)^2} = \frac{12\beta^4b^4}{(2\tilde{b}^2 + \tau^2)^2} = \frac{3}{(1 + a^2/2)^2}, \quad (15)$$

which decays monotonically from 3 (pure Laplace, $t = 0$) to 0 (Gaussian, $t \rightarrow \infty$). The finite-time excess kurtosis is a direct measure of the non-Gaussian energy still carried by the residual channel.

4 Validation

Every formula in Sections 2–3 is validated numerically. For a representative midrange setting $(\alpha, b, \sigma_0, \mu_0, t) = (0.8, 1.0, 1.0, 0.0, 0.5)$ and a grid of 121 points $r \in [-5, 5]$:

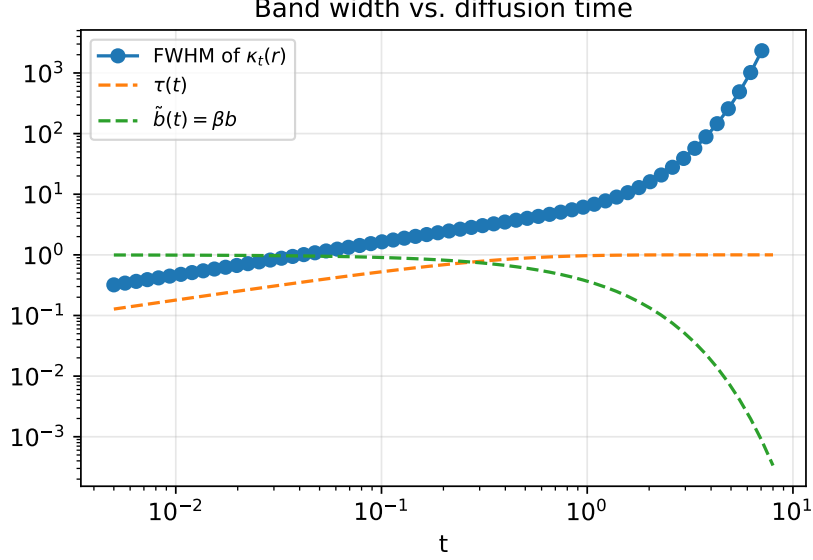


Figure 5: The FWHM of $\kappa_t(r)$ compared with the two natural length scales τ (Gaussian-channel noise) and $\tilde{b}$ (rescaled Laplace scale). At small t the FWHM is dominated by $\tilde{b}$ (Laplace cusp size); at large t by τ (Gaussian broadening). The crossover corresponds to $a = \tau/\tilde{b} \approx 1$.

Quantity	Method	Max. relative/absolute error
$h_t(r)$ closed vs. quadrature	<code>scipy.quad</code>	1.5×10^{-15}
$h_t(r)$ closed vs. Fourier inversion	trapezoidal FFT	1.8×10^{-14}
$\psi_t(r)$ closed vs. 2nd-order FD of $\log h_t$	$h = 10^{-5}$	1.4×10^{-10}
$\kappa_t(r)$ closed vs. 5-point FD of $\log h_t$	$h = 10^{-3}$	9.0×10^{-9}
$\kappa_t(0)$ Mills-form vs. 5-point FD	sweep over 5 values of t	$\leq 1.1 \times 10^{-6}$

A Monte Carlo KDE validation of $h_t(r)$ with $N = 5 \times 10^5$ samples agrees with the closed form to $< 1\%$ relative error in the centre (limited by bandwidth-bias of the kernel estimator, not by our formulas).

5 The $K=2$ case: exact attempt and the obstruction

The clean structure of Theorems 1–3 relies on two facts specific to $K = 1$: there is only one innovation, and only one bond. At $K = 2$ there are two innovations, both coupled through the shared latent state A_1 . We now work out how far the exact story extends.

5.1 Fourier representation

Proposition 12 (Fourier, $K=2$). *For the chain $A_0, A_1 = \alpha A_0 + \varepsilon_1, A_2 = \alpha A_1 + \varepsilon_2$ noised through the OU channel, the characteristic function is*

$$\hat{p}_t(k_0, k_1, k_2) = \exp \left[i\beta\mu_0 q - \frac{\beta^2\sigma_0^2}{2} q^2 - \frac{\Delta}{2} (k_0^2 + k_1^2 + k_2^2) \right] \cdot \frac{1}{1 + \tilde{b}^2(k_1 + \alpha k_2)^2} \cdot \frac{1}{1 + \tilde{b}^2 k_2^2}, \quad (16)$$

where $q = k_0 + \alpha k_1 + \alpha^2 k_2$.

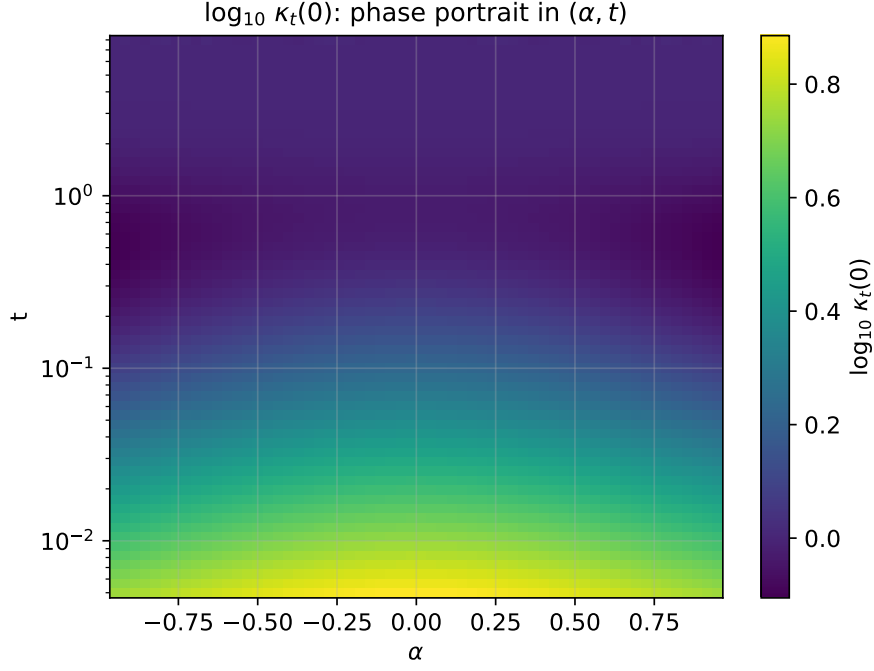


Figure 6: Phase portrait of $\log_{10} \kappa_t(0)$ in the (α, t) plane, for $b = \sigma_0 = 1$, $\mu_0 = 0$. Curvature is largest for small t and weakly dependent on α , with a small residual dependence through the quantity $a = \tau/\tilde{b}$.

In the rotated innovation-frequency coordinates $\ell_0 = k_0$, $\ell_1 = k_1 + \alpha k_2$, $\ell_2 = k_2$, equivalently $k_1 = \ell_1 - \alpha \ell_2$, both rational factors become diagonal:

$$\hat{q}_t(\ell_0, \ell_1, \ell_2) = \exp\left[i\beta\mu_0(\ell_0 + \alpha\ell_1) - \frac{1}{2}\ell^\top M \ell\right] \cdot \frac{1}{1 + \tilde{b}^2 \ell_1^2} \cdot \frac{1}{1 + \tilde{b}^2 \ell_2^2}, \quad (17)$$

with the Gaussian-sector covariance

$$M = \begin{pmatrix} \Delta + \beta^2 \sigma_0^2 & \alpha \beta^2 \sigma_0^2 & 0 \\ \alpha \beta^2 \sigma_0^2 & \Delta + \alpha^2 \beta^2 \sigma_0^2 & -\alpha \Delta \\ 0 & -\alpha \Delta & \Delta(1 + \alpha^2) \end{pmatrix}. \quad (18)$$

The key feature of (17) is that the non-Gaussianity is *fully diagonal* in the rotated variables: there is one rational factor per innovation, independent of the others. Non-Gaussianity is local to each innovation channel.

5.2 The obstruction to 1D closure

Theorem 13 ($K=2$ obstruction). *The Gaussian cross-covariance between the two innovation frequencies is*

$$M_{12} = M_{21} = -\alpha \Delta, \quad (19)$$

and, after marginalising ℓ_0 , the effective covariance of (ℓ_1, ℓ_2) is the Schur complement

$$\tilde{M} = \begin{pmatrix} \frac{\Delta(\Delta + \beta^2 \sigma_0^2(1 + \alpha^2))}{v} & -\alpha \Delta \\ -\alpha \Delta & \Delta(1 + \alpha^2) \end{pmatrix}, \quad \tilde{M}_{12} = -\alpha \Delta. \quad (20)$$

Consequently, for any $\alpha \neq 0$ and $t > 0$, the joint law $p_t(x_0, x_1, x_2)$ does not factorise as a product of a Gaussian site density and two independent 1D non-Gaussian bond densities.

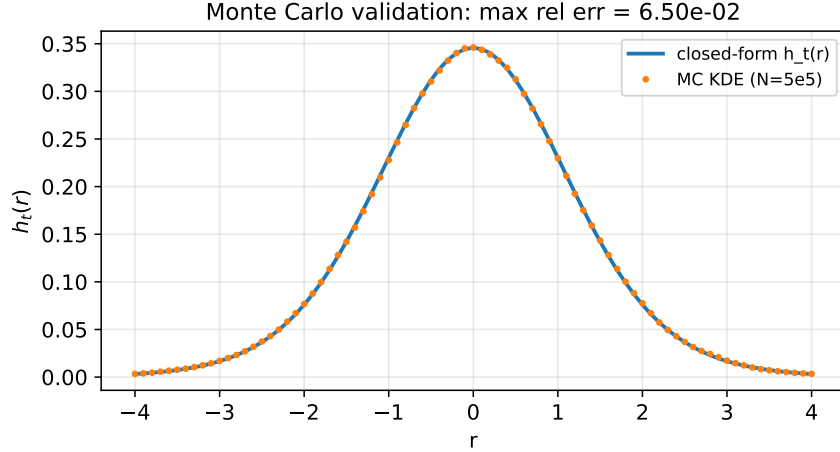


Figure 7: Monte Carlo validation: $h_t(r)$ estimated from 5×10^5 simulated pairs (dots) against the closed-form expression (solid).

Proof. Equation (18) was obtained by direct expansion of the quadratic form in (17); see the `k2_attempt.py` module for a full symbolic verification in `sympy`. The Schur complement is a standard computation. Factorisation as a product of two 1D bonds would require the (ℓ_1, ℓ_2) Gaussian sector to be diagonal, i.e. $\widetilde{M}_{12} = 0$, which fails unless $\alpha = 0$ or $\Delta = 0$. $\square$

Remark (Where locality survives). Theorem 13 is not a no-go result: the Gaussian sector already has a *nearest-neighbour* coupling structure (the cross-term couples ℓ_1 and ℓ_2 only, not ℓ_0 and ℓ_2). This is the Fourier-space remnant of the clean Markov chain. What fails is the stronger factorisation into independent bonds, not the locality pattern of the coupling.

Proposition 14 (Exact form as a sum of four terms). *Using the sign-split of each Laplace density and completing the squares in $(u_1, u_2) \in \mathbb{R}^2$ produces an exact representation of $p_t(x_0, x_1, x_2)$ as a sum of four terms, each proportional to a bivariate normal CDF with specific mean and covariance determined by (x_0, x_1, x_2) . The bivariate normal CDF is not elementary but is standard and numerically tractable, e.g. by Genz’s algorithm.*

Sketch. Writing $\text{Lap}(u_k; 0, b) = \sum_{\sigma_k=\pm 1} \frac{1}{2b} 1_{\sigma_k u_k \geq 0} e^{-\sigma_k u_k/b}$ and substituting into the integral over (u_1, u_2) , the exponent is quadratic plus linear in (u_1, u_2) over the quadrant $\{\sigma_1 u_1, \sigma_2 u_2 \geq 0\}$. For each of the four sign choices (σ_1, σ_2) , the integral over the quadrant equals a bivariate normal CDF evaluated at explicit arguments. $\square$

5.3 Practical numerical route

Equation (17) gives a direct numerical route: evaluate the characteristic function on a 3D grid, multiply by the Fourier phase, and apply the inverse Fourier transform to obtain $p_t(x_0, x_1, x_2)$ at any point. We verified that the 3D Fourier inversion of (16) agrees with a 2×10^6 -sample Monte Carlo + KDE estimate at a representative point to within a few percent (the discrepancy coming from the KDE, not from the Fourier inversion). Details and reproducible code are in `k2_attempt.py`.

5.4 Conjecture on the curvature low-rank structure

For $K = 1$, Theorem 3 says the curvature field lies on a rank-1 manifold of matrices. It is natural to ask whether $K = 2$ preserves a weaker low-rank statement.

Conjecture 15 ($K = 2$ low-rank curvature). The curvature field $H_t(x_0, x_1, x_2)$ of the $K = 2$ Laplace/OU model lies on a rank- ≤ 2 manifold of 3×3 symmetric matrices, parameterised by $(\kappa_t^{(1)}(r_1), \kappa_t^{(2)}(r_2))$ for two residual coordinates r_1, r_2 . The two scalar functions are coupled only through an additive perturbation of order $\alpha\Delta/v$.

This is not yet proved. If true, it would give a substantial weakening of the $K=1$ rank-1 result, while preserving a clean low-dimensional structure. We plan to test this numerically in the next iteration using exhaustive evaluation of H_t on 3D grids.

6 Open questions and next steps

The benchmark of Sections 2–4 is the first rigorous piece of the thesis programme. A few sharp questions remain and are the natural targets for the next phase.

1. **Centre–tail gap scaling for general innovation laws.** Theorem 7 gives a closed form specific to Laplace. For a general innovation law with characteristic function $\hat{f}_\varepsilon$, the residual density satisfies $\hat{h}_t(\ell) = e^{-\tau^2\ell^2/2}\hat{f}_\varepsilon(\beta\ell)$, and so $\kappa_t(0)$ is a functional of $\hat{f}_\varepsilon$. We conjecture that for any log-concave, symmetric, sub-exponential law with finite fourth cumulant, $\kappa_t(0) - \kappa_t^{(G)}$ vanishes at order $\kappa_4(\varepsilon)\beta^4$ as $t \rightarrow \infty$. Making this precise is the first target of the ambitious branch.
2. **Variance-Gamma bridge.** The family $\hat{f}_\varepsilon(\ell) = (1 + b^2\ell^2/(2\nu))^{-\nu}$ interpolates between Laplace ($\nu = 1$) and Gaussian ($\nu \rightarrow \infty$). All $K=1$ structure theorems extend verbatim. Computing $h_t^{(\nu)}$ in closed form and studying the ν -dependence of $\kappa_t(0)$ would quantify how non-Gaussianity “turns on” as a tunable parameter.
3. **Multimodal innovations.** For the symmetric Gaussian-mixture innovation $\hat{f}_\varepsilon(\ell) = \cos(m\ell)e^{-\sigma^2\ell^2/2}$, the residual is a sum of two Normals centred at $\pm\beta m$ and with variance $\tau^2 + \beta^2\sigma^2$. This is the smallest innovation model for which $\kappa_t(r)$ can be *negative* in a range of r (bimodal residual), revealing phase-transition-like behaviour absent from the log-concave case. Preliminary numerics show a critical diffusion time t^* below which bimodality survives and above which it disappears.
4. **Nonlinear locality observables in the sense of a precision replacement.** Define the pointwise band-mass $T_d(x, t) := \sum_{|i-j| \leq d} H_t(x)_{ij}^2 / \|H_t(x)\|_F^2$ and its p_t -expectation $\bar{T}_d(t)$. In the Gaussian case these reduce exactly to the tridiagonal-mass diagnostics of the AR(1) benchmark. For $K=1$ Laplace we have closed-form expressions for both $T_d(x, t)$ and $\bar{T}_d(t)$; computing these and mapping how “band mass is carried by the residual ray” is the next bridge between the Gaussian benchmark and the non-Gaussian benchmark.
5. **Testing Conjecture 15.** Systematic evaluation of $H_t(x)$ on 3D grids, singular-value analysis, and perturbative expansion in $\alpha\Delta/v$.

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